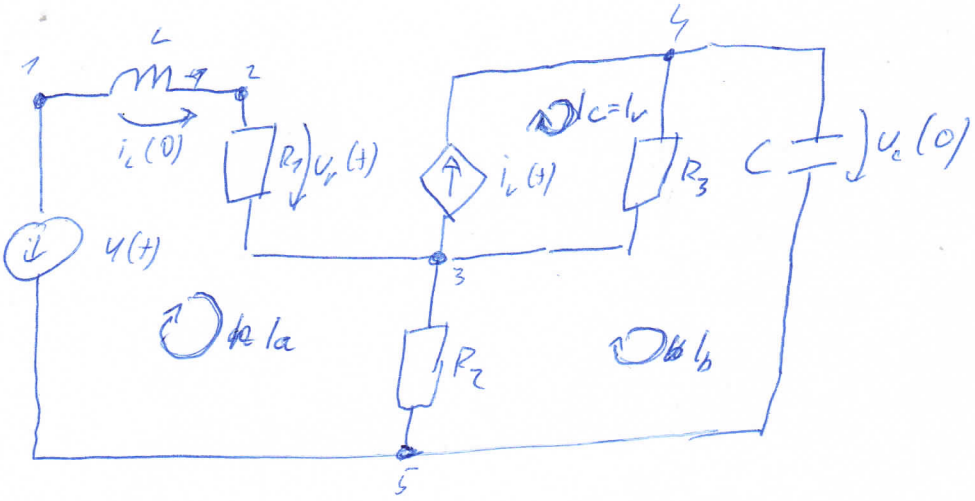
5

tysela



$i_v(t) = G u_r(t)$

$MUN = 5 - 1 - 1 = 3$

$MSP = 3 - 1 = 2$

$$-U + R_2(i_a - i_b) + R_1 i_a + L\frac{di_a}{dt} = 0$$

$$\frac{1}{C}\int_0^t i_b \, d\tau - u_c(0) + R_2(i_b - i_a) + R_3(i_b - i_v) = 0 \; ; \; i_v = R_1 i_a \mapsto$$

$$\cancel{-U + R_2(i_a(t) - i_b(t)) + R_1 i_a(t) + L\frac{di_a(t)}{dt} = 0}$$

$$\cancel{\frac{1}{C}\int_0^t i_b(t)\,d\tau - u_c(0) + R_2(i_b(t) - i_a(t)) + R_3(i_b(t) - R_1 i_a(t)) = 0}$$

$$\boxed{\begin{aligned} &-U + R_2(i_a(t) - i_b(t)) + R_1 i_a(t) + L\frac{di_a(t)}{dt} = 0 \\ &\frac{1}{C}\int_0^t i_b(t)\,d\tau - u_c(0) + R_2(i_b(t) - i_a(t)) + R_3(i_b(t) - R_1 i_a(t)\cdot G) = 0 \end{aligned}}$$